PRISM: Property Regression and Interpretable Structures for Molecules

Misael Andre D. Maningo
College of Engineering
University of the Philippines Diliman
Quezon City, Philippines
mdmaningo@up.edu.ph

Abstract*—Artificial intelligence has increasingly been leveraged in computer-aided drug design, particularly for predicting critical physicochemical properties. However, many deep learning architectures function as "black boxes," obscuring the specific atomic contributions driving the final predictions. To address this limitation, this study introduces the Property Regression and Interpretable Structures for Molecules (PRISM) framework, which maps molecular graphs to an artificial neural network utilizing Graph Attention Networks (GAT). The framework is evaluated on predicting the logarithm of the partition coefficient (logP), a key determinant of lipophilicity and subsequent drug bioavailability. To achieve model transparency, we integrate three explainability methodologies: Integrated Gradients, SHAP, and intrinsic attention mechanisms. The resulting model, LogPGNN, achieved high performance on the computational ZINC 250k dataset and successfully captured the underlying rule-based heuristics of the Crippen method. Conversely, when trained on the empirical SangsterLogP dataset, LogPGNN exhibited mediocre performance, a limitation likely attributable to the structural variance introduced by scaffold splitting and the more complex chemical interactions inherent to the experimental data. Nevertheless, the explainability visualizations revealed consistent, chemically intuitive atom-level attributions for the total logP. These findings demonstrate that the PRISM framework holds significant potential for expansion into multi-property prediction, ultimately empowering researchers to perform targeted, atom-level modifications in rational drug design.

***Index Terms*—molecular graphs, feature attribution, logP**

I. Introduction

Drug design is the process of developing effective and safe molecules that can prevent or treat diseases [1]. Computer-aided drug design makes the process more cost and time-efficient by proceeding with synthesis and clinical trials only after potential drug candidates have been identified through methods such as virtual screening [2]. Current methods in drug design have seen the rise of AI methods and implementations [1], as seen in the AI platform Expasy by the Swiss Institute of Bioinformatics [3] and AlphaFold by Google [4].

Deep learning (DL) models, which involve multiple layers of artificial neural networks (ANNs), have an inherent "black-box" problem, making their final predictions for any property less interpretable [5]. This reduces the trust in and interpretability of the DL model, hindering progress in drug design. Explainable artificial intelligence (XAI) is a method that solves this problem by explicitly attributing causes of the model's output, i.e., through feature attribution or feature importance [5]. XAI methods includes SHapley Additive exPlanations (SHAP) values [6] and integrated gradients [7].

Besides the effectiveness of drugs, they must also reach the target location within the body in the first place. This is related to the field of pharmacokinetics, which focuses on the steps of Absorption, Distribution, Metabolism, and Excretion (ADME) [8]. One of the important chemical properties is its lipophilicity, which is measured as the logarithm of the partition coefficient or logP [9]. A higher value corresponds to fat-soluble, while a lower value (going to negatives) corresponds to water-soluble. Since the body has environments with both extremes, most drugs, in general, are preferred to have moderate values, balancing both properties.

The objective of this project is thus to design the PRISM (Property Regression and Interpretable Structures for Molecules) framework to enable chemical property DL regression utilizing XAI methods. Specifically, to test the viability of the framework, only logP will be used for the regression of drugs from a drug database with known values. The DL model should also give satisfactory performance and exhibit explainability.

II. Related Studies

A. Graph Neural Network

A group of connected objects is called a graph [10]. Specifically, it consists of nodes (or vertices) that are connected by edges. There is further classification on whether these edges form an order (directed graphs) or not. Some data are better represented as graphs, such as social networks, where people are nodes and social connections are edges. Another notable example is that of molecules where each atom acts as a node and chemical bonds as edges.

A graph neural network (GNN) is a type of ANN that takes an input graph for prediction [10]. A message-passing mechanism ensures that information gets passed and aggregated between nodes through their connected edges before being processed further. GNN can focus on predicting specific values for each of the nodes, edges, or the entire graph. Generally, the structure of a GNN is composed of a message passing convolution layer, followed by the output linear layer for the final prediction. Types of GNN include the basic Graph Convolution Networks (GCN), which utilize convolutions for message-passing, and more advanced methods such as Graph Attention Networks (GAT), which utilize the attention mechanism.

The attention mechanism was invented by Vaswani et al. [11], which is colloquially found in modern large language models (LLMs), specifically with its transformer ANN architecture. The attention mechanism basically allows the model to mimic human attention by prioritizing certain relationships between features. In the case of LLMs, it allows the model to prioritize the relationship between words, which is crucial for language context. The original GAT developed by Veličković et al. [12] utilizes this same attention mechanism for GNNs, which prioritizes certain connections in message passing compared to the equal priority in normal GCNs. Brody et al. [13] developed GATv2, which improves on the original GAT by utilizing dynamic attention, which leads to better performance.

B. Feature attribution

Feature attribution is an XAI method that seeks to find the specific numerical contribution of each input feature to the final output of a function/model [7]. Two mathematical axioms are required for a robust feature attribution: sensitivity and feature invariance. *Sensitivity* states that if one feature changes while the others stay constant, causing a change in the output prediction, then that specific feature should have a non-zero attribution. *Feature invariance*, on the other hand, states that if two potential models have the same input-output pairs, i.e., functionally equivalent, then the corresponding attribution is also equivalent, regardless of the model implementations. All of these feature attributions are with respect to a baseline input, which is generally chosen by picking the input that outputs a zero prediction.

Sundararajan et al. [7] developed *integrated gradients (IG)* as a more sophisticated way to get the corresponding feature attribution, especially in neural networks, where naive methods, such as using only gradients, break some of the given axioms. IG also has the property of *completeness* in which summing the various attributions for each input feature is equal to the final prediction. Part of the process of integrated gradients is approximating an integral function from the baseline to the actual input using numerical summation. Since DL methods are differentiable, IG is very useful for them, and the authors have shown various applications of ANNs across many fields, including object recognition, diabetes prediction, question classification, translation, and even chemistry models.

While IG is path-based, *SHAP* is another feature attribution method that uses additive feature attribution based on game theory [6]. This attribution framework has analogous axioms as were discussed before: consistency, missingness, and local accuracy. SHAP works by analyzing how the output changes when various combinations of features are used as a “baseline” feature, then averaging the marginal contribution of each feature. SHAP has the advantage over IG of being more model-agnostic, as it can be used with more AI models beyond DL. However, it is computationally expensive to compute as more features are involved due to a combinatorial explosion, and thus, approximate methods such as SHAP sampling are used.

C. logP

Besides other important drug properties such as QED and SAS, logP is an important chemical property for drugs due to bioavailability and is thus used as an effective screening [9]. LogP is defined with the following equation:

$$\log \mathrm{P} = \log_{10}\left(\frac{[\mathrm{organic}]}{[\mathrm{aqueous}]}\right) \tag{1}$$

where the brackets [ ] indicate the concentration of the solute drug in an organic (typically n-octanol) and aqueous (typically water) partition. A more positive value implies lipophilic (fat-soluble), whereas a more negative value implies hydrophilic (water-soluble).

Lipinski’s Rule of 5 (Ro5) is a useful rule of thumb in filtering out chemicals based on the various chemical properties [9, 14]. In terms of logP, it states that its value should not be greater than 5, i.e., $\log \mathrm{P} < 5$. A very high logP value means very fat-soluble, which interferes with bioavailability and may even accumulate in fatty tissues, leading to toxicity. Note, however, that this rule is in general cases, and certain body parts may actually prefer higher logP values, such as for sub-lingual absorption.

III. Methodology

A. Software & Hardware

All coding was done on Jupyter Notebook with Python 3.12.13. Common Python packages were used, besides the special packages that will be mentioned correspondingly.

All code was run in the University of the Philippines, College of Engineering’s SHARC HPC, which contains an AMD EPYC 77421 CPU with TB of RAM and eight NVIDIA A100-SXM4 GPUs with 40 GB of VRAM each.

For reproducibility, the random seed was set to 42 for various steps with randomization. However, due to complexity, this was not implemented for the PyTorch training.

B. Dataset

There are two datasets that were used: ZINC 250k and the Sangster LogP dataset.

The ZINC 250k is a subset of the very large ZINC database, which currently contains over 37 billion molecules in 2D format and around 4.5 billion molecules in ready-to-dock 3D formats [15, 16] . Due to its size, it is useful in drug screening tasks. Specifically, this dataset contains 249,455 molecules. The 2D format for the molecules is SMILES (Simplified Molecular Input Line Entry System), which is a chemical string notation with its own semantics and syntax (e.g., benzene is ‘c1ccccc1’) [17]. Another advantage of this dataset is the inclusion of 3 chemical properties for each molecule: logP, QED, and SAS. However, note that these properties were calculated computationally based on an algorithm. Specifically for logP, they utilized an additive rule-based method called the Crippen method in approximating the logP [18]. This means that the feature attribution is expected to capture the Crippen method rules rather than the actual chemistry. Thus, we use this dataset as a "sanity check" to determine whether the PRISM framework attributes the contribution of each atom in the same way as the Crippen method.

The SangsterLogP dataset is claimed by the authors to be the "largest publicly available dataset of logP values", and for emphasis, "experimental" logP values [19]. This dataset, containing exactly 23,520 molecules, was meticulously curated by the authors using various filtering techniques that balance the dataset size with the accuracy of each measurement. This dataset then tests whether the PRISM framework can capture the more complex chemical interactions of the molecule that yield its actual experimental logP.

For splitting the ZINC 250k data, a random 70-10-20 split was done for training, validation, and testing sets, respectively. On the other hand, the same partition percentage was done on the SangsterLogP dataset, but using scaffold splitting. Scaffold splitting groups molecules with the same core structure on the same subset to mimic the reality of virtual screening, where molecules are structurally distinct [20]. In general, this increases the generalization of the model at the cost of harder prediction due to extrapolation. The Python package called scikit-fingerprints has a function called `scaffold_train_valid_test_split` that implements this scaffold splitting [21].

To see the difference between the three sets, the KDE of the following properties is graphed per set: logP, molecular weight (MW), H-bond acceptors (HBA), H-bond donors (HBD), topological surface area (TPSA), and rotatable bonds. These properties are calculated computationally by the RDKit package [22]. Also, a t-SNE was implemented to show the distribution of each set.

C. Preprocessing

The SMILES string for each molecule was converted to a graph in the form of PyTorch Geometric (PyG) `Data` object using the RDKit package [23]. Shown in Table I are the atomic (node) and bond (edge) information that were calculated by RDKit and embedded into each node/edge for the entire molecular graph.

TABLE I
ATOMIC (NODE) AND BOND (EDGE) FEATURES FOR EACH MOLECULAR GRAPH.

Atom (node) properties	Bond (edge) properties
Atomic number	Bond Type
Degree	Stereochemistry
Total Number of Hs	Conjugation State
Total Valence	
Formal Charge	
Gasteiger Charge	
Hybridization	
Aromaticity	
Ring	
Chirality	

D. Model Architecture

The ANN implementation was done using the PyTorch [24] and PyG package. The input graph is fed to a `GATv2Conv` layer followed by a `BatchNorm` layer (for stabilization and accelerating training) and finally the `ReLU` activation layer. These layers are repeated a number of times (a hyperparameter) but with the addition of a residual addition at the end to prevent oversmoothing of GNN, which degrades prediction performance [25]. After those, a global mean pooling and global max pooling, which collapse the graph into two vectors that are finally concatenated. Finally, these are fed to a simple three FFN or linear layer with ReLU activation (except the last) to produce the single logP prediction. To discourage overfitting, a Dropout layer was implemented after each linear layer except the last with $p = 0.4$. We call this GNN model that predicts logP the "LogPGNN" model.

The loss to be minimized is the mean squared error (MSE) using the AdamW solver [26]. For the final training of the model, it was done for 100 epochs with a batch size of 1024.

The hyperparameters of the model were tuned using Optuna [27]. Table II shows the various hyperparameters that were tuned from the GAT layers, the linear layers, and finally the AdamW solver. Specifically, a Bayesian optimization using the TPE (Tree-structured Parzen Estimator) algorithm was utilized to find the best hyperparameters that minimized the validation set error after training 32 epochs.

TABLE II
HYPERPARAMETERS TO TUNE FOR THE MODEL.

Hyperparameter	Values
GAT `num_conv`	1, 2, 3, 4, 5, 6
GAT `hidden_dim`	32, 64, 128, 256
GAT `num_heads`	2, 4, 6, 8
Linear `funnel_dim`	16, 32, 64
AdamW `learning_rate`	(1e-4, 1e-2) (log)
AdamW `weight_decay`	(1e-3, 1e-2) (log)

Note that this hyperparameter tuning was only done for the model trained on SangsterLogP and not for ZINC 250k. Instead, default values were used.

The final training of the model that used SangsterLogP was repeated 5 times to assess training stability, especially given the randomness of the PyTorch layers, as noted earlier. However, only the first trained model was used in testing and explainability.

E. Testing

To test the accuracy of the model, regression metrics such as R^2 were calculated.

With respect to Lipinski's Ro5, which states that drugs should have logP less than 5, one can also translate this into a classification problem by predicting if the drug passes or violates the rule (violating being called a "beyond Ro5 drug or bRo5). Thus, classification metrics would also be measured, including a confusion matrix.

For the feature attribution, both integrated gradients and SHAP would be used to identify and highlight which atoms contributed to the final logP prediction of the entire molecule. Specifically, the integrated gradient and sampling SHAP implementation of the package Captum was used [28]. The "baseline" was calculated by averaging the properties of all atoms of all molecules in each dataset. The results of this feature attribution are compared to a Crippen method attribution for comparison.

Extracting the attention weights of the last GAT layer shows which edges (or bonds) are important in exchanging information during message-passing. Since the attention of an atom A with atom B is not necessarily the same as the reverse attention, these attention weights were combined by taking the average.

The atom and bond highlighting is then drawn using RDKit. A divergent color map is used for the atom attribution, with reddish colors representing negative logP contribution, green/blue/purple colors representing positive logP contribution, and no highlight colors representing no contribution. For the edge attribution, the color scales from transparent to dark green, corresponding to no attention to high attention weights.

Note that the hydrogen atoms that are not explicitly drawn in a skeletal diagram of a molecule also contribute to the logP. However, since they will clutter the diagram, they are omitted, and their corresponding contribution are passed on to the heavy (i.e, non-hydrogen) atom that they are attached to.

IV. Discussion and Analysis of Results

A. Dataset splitting

Shown in Figure 1 are the distributions of 6 chemical properties for each of the training, validation, and testing sets for the SangsterLogP dataset, along with the corresponding 2D t-SNE visualization. Visually, with the distribution curves, one can clearly see a distinction between the subsets. In using an independent t-test between the training and test set, it was calculated that all p-values for all 6 properties were virtually 0, which implies that the hypothetical population means between the two subsets are statistically different. This means that the scaffold folding was successful in splitting up different groups to each of the subsets and thus generalization would be stronger but harder to achieve [20]. However, do note that the very low p-value is likely due to the very large sample size, whereas the actual difference of the properties or the effect size is insignificant, which might imply that the data in each set should still be in some ways "scientifically" similar [29].

This somewhat similarity is further seen in the 2D t-SNE plot of the 6 properties in Figure 1 as both the validation and test points are distributed where the training set points are, i.e., there are no regions where only one subset is there exclusively. Note that this analysis is qualitative, based on visually inspecting the t-SNE diagram. This similarity implies that although the molecules in each subset have different overall structures and different property distributions, taking them altogether, the molecules still explore somewhat the same chemical space and thus generalization can be expected.

Although not shown here, it was statistically calculated that for the random splitting for the ZINC 250k dataset, there were no significant differences between the training and test set, proving that a random splitting leaks data from the test set to the training set [20].

B. Hyperparameter tuning

The best hyperparameters for the model trained on SangsterLogP are shown in Table III. As expected, the hyperparameter values chosen are not the extremes given in the range: not too small that the model will underfit, but not too large that it will overfit, especially with scaffold splitting, where generalization is a must.

TABLE III
Best hyperparmeters.

Hyperparameter	**Values**
GAT `num_conv`	4
GAT `hidden_dim`	128
GAT `num_heads`	4
Linear `funnel_dim`	16
AdamW `learning_rate`	0.0007
AdamW `weight_decay`	0.0541

Optuna also gives which hyperparameters have the most influence or importance in minimizing the validation loss. In this case, it is unsurprising that the AdamW learning rate has the largest importance since it accelerates the training of the 30 epochs only that were given to each model for hyperparameter tuning. The second important hyperparameter was the hidden dimensions of all GATv2Conv layers, as it is the majority of all layers in the model. The third is the AdamW weight decay, which is a form of regularization that increases generalization [26]. The rest of the hyperparameters were deemed significantly less important.

C. Training

Figure 2 shows the loss curve of the first trained model on SangsterLogP, which, highlighted in red, suffered some form of overfitting. This is to be expected again due to the nature of scaffold splitting, but it is implied that this overfitting would have been worse without the regularization techniques utilized, such as the Dropout layers and the weight decay.

Also, Figure 2 shows that the MSE loss is more or less stable during training as seen by the relatively small standard deviation. It is expected that the validation loss has the smallest variance since the hyperparameters were optimized using said validation set.

Although not shown here, the model trained on ZINC 250k has no noticeable overfitting.

D. Regression

The model trained on ZINC 250k shows promising regression performance as shown in the parity plot in Figure 3 with very high $R^2 = 0.966$ and a relatively low mean absolute error (MAE) and root mean square error (RMSE). This good

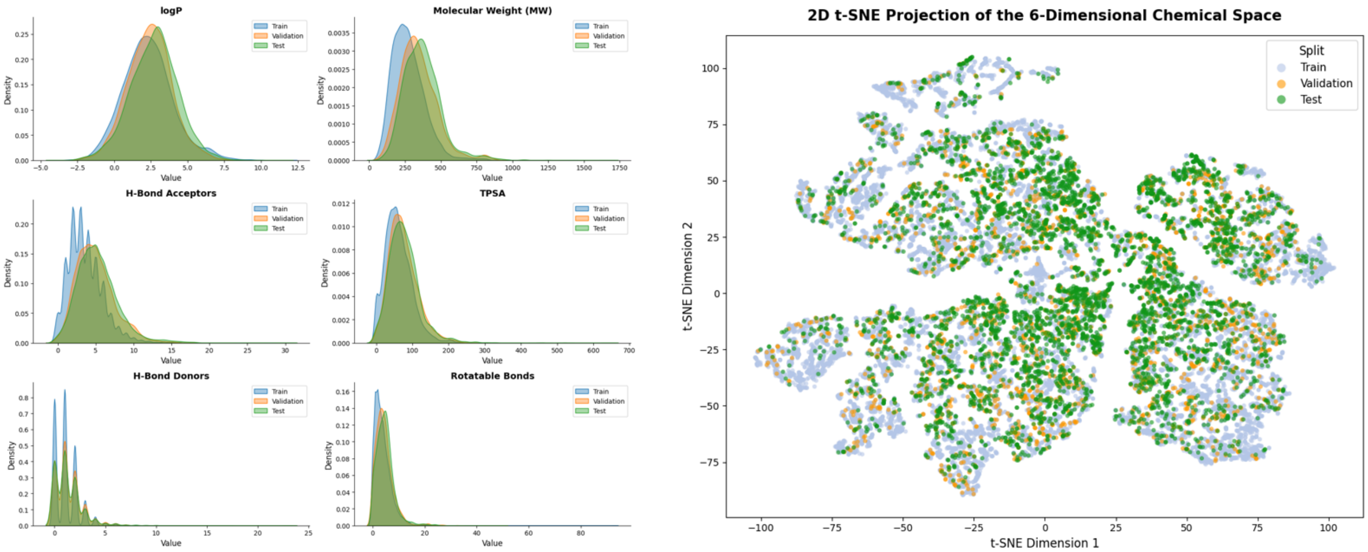


Fig. 1. Chemical property distribution (left) and t-SNE (right) of the subsets for SangsterLogP.

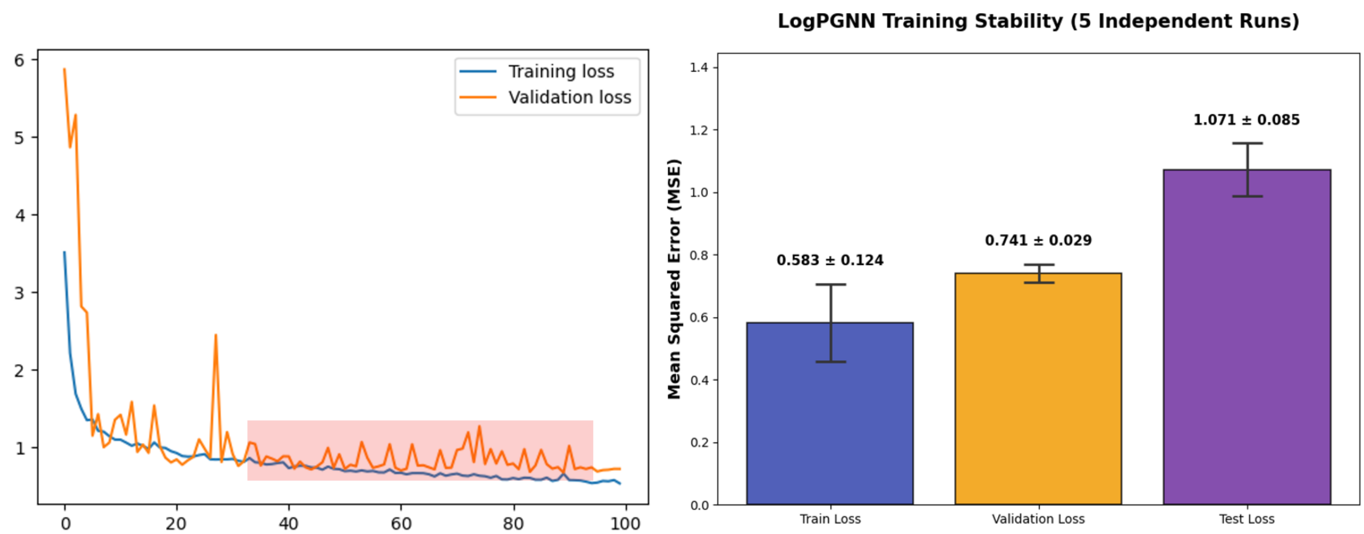


Fig. 2. Left: Training loss curve of the first training model on SangsterLogP. Right: MSE Loss bar plot with error bar (1 standard deviation) for each subset on SansgerLogP.

performance is to be expected since the model is basically interpolating due to the random splitting. Also, since there are more datapoints on the ZINC 250k, the model was able to be trained on more samples, which leads to more predictive performance.

However, the regression performance on the SangsterLogP (shown in Figure 3) is mediocre with $R^2 = 0.598$ and a relatively large MAE and RMSE. But it is to be remembered that SangsterLogP is sort of extrapolating due to the nature of scaffold splitting, so that this mediocre result may be good enough. Also, although not shown, comparing the parity plot in using the Crippen method to estimate the logP values for the test set of the SangsterLogP dataset gives an $R^2 = 0.425$ and a slightly worse MAE and RMSE. This means that, especially for this specific dataset, the trained model actually performed better than the rules-based Crippen method. This means that the model captured the more complex and real chemical interactions for the final logP than what was attributed by static rules in the Crippen method [18].

Also, it was done earlier in experimentation that using a random splitting of the SangsterLogP dataset actually yielded better regression metrics. However, due to the generalization provided by scaffold folding, which mimics real-life virtual screening, using the scaffold splitting was utilized for the final results.

Although a study by Guo et al. [20] showed that scaffold splitting can overestimate the performance, using scaffold splitting is still used in this experiment since it is a standard

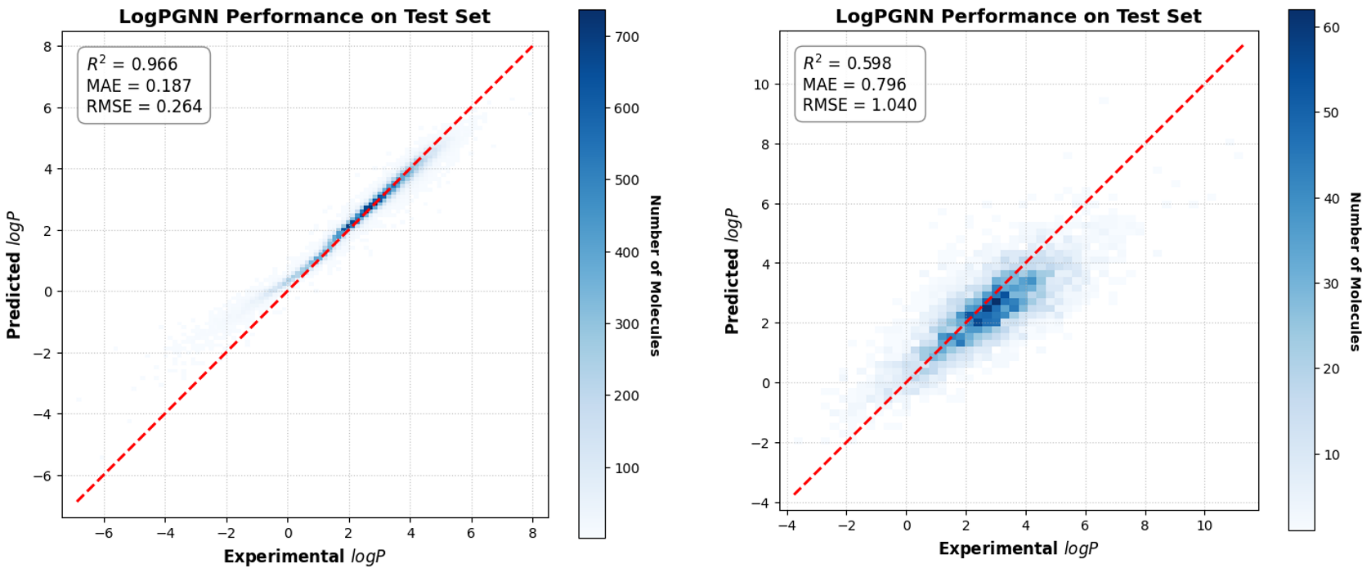


Fig. 3. Parity plot curve of the models trained on ZINC 250k (left) and SangsterLogP (right) dataset.

used and the performance of the model is not the main objective (but rather the feature attribution).

E. Classification

Shown in Figure 4 are the confusion matrices of the models trained on the ZINC 250k and SangsterLogP dataset in classifying if a molecule passes Lipinski's Ro5 criteria, that it has a logP less than 5. Although both naively give a very high accuracy (99% and 94%), they actually have a very low recall for the FAIL or bRo5 molecules (31% and 18%), which results in a low average macro F1-score (73% and 63%) and low Matthews correlation coefficient (MCC) (0.5307 and 0.3404).

This low recall is likely due to the fact that bRo5 molecules are rare in both datasets (1.60% and 6.44%) and thus the model is biased to predict that most molecules have logP less than 5. It was also checked that both random and scaffold splitting have more or less similar percentages of bRo5 in each subset, so splitting is not the reason for poor performance, but rather this is just a classic case of severe class imbalance of the entire dataset.

This low recall of bRo5 molecules implies that the model is more prone to false positives of molecules that pass this logP criteria. A false positive in drug design is more costly since a lot of time and synthesis are used to create a drug that might not actually work. Thus, one might "sacrifice" some of the precision to increase the recall by shifting the logP threshold of the model of classifying a bRo5 molecule. This tradeoff can be seen with the precision-recall graph as shown in Figure 5 for the model trained on SangsterLogP. Comparing the area under the curve (AUC) of this PR-curve with the random baseline shows that a tradeoff is somewhat worth it.

However, do note that there are some useful drugs that violate some criteria in Lipinski's rule, and some studies actually support that bRo5 drugs might be useful in some cases and should be explored more, such as with Proteolysis Targeting Chimeras (PROTACs) [30, 31].

F. Explainability

1) ZINC 250k: Shown in Figure 6 are the feature attribution of the logP on some random test set molecules. Specifically, the Crippen method attribution was compared with that of the attribution by integrated gradients of the LogPGNN model. Note that the logP values between the two methods are relatively close, and thus we can directly compare their attributions. Also, it is to be remembered that the ZINC 250k ground truth is using the Crippen method.

It can be seen visually that the LogPGNN model trained on ZINC 250k captures most of the Crippen method rules as described by Wildman and Crippen [18]. Notably, it correctly attributed the large negative magnitude contribution of the protonated amines (NH_3^+-, NH_2^+- and NH^+-) of the first 3 molecules in Figure 6 as shown by the deep red colors, as this ionic molecule is more soluble in aqueous solution. It was also attributed that halogens (fluorine, chlorine, bromine, and iodine) consistently increase the logP as halogenation in general increases lipophilicity [32].

However, there are some discrepancies which are indicated by the red circles, most of them involving an opposite sign of logP contribution (i.e., from positive to negative logP attribution and vice versa). For example, in the sulfonyl group (-SO_2-), the Crippen method assigns negative logP attribution to both oxygen atoms and barely any contribution of the sulfur atom, whereas the LogPGNN model reverses the contributions, where sulfur has the most influence in decreasing logP, while the oxygens do not contribute anything. The urea group (carbonyl C=O with two amide NH groups) of molecules 4 and 5 also shows this switch in sign of the influence with the oxygen and nitrogen atoms. Also, the sign of the attribution is flipped for the carbonyl group (C=O) as seen in molecule

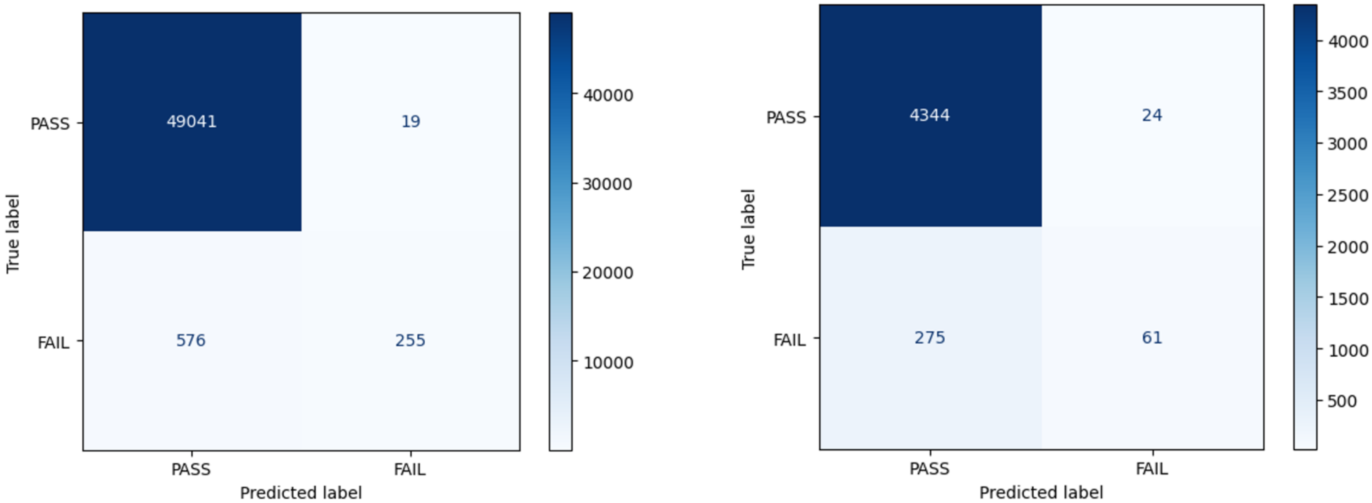


Fig. 4. Confusion matrix of the model trained with ZINC 250k (left) and SangsterLogP (right).

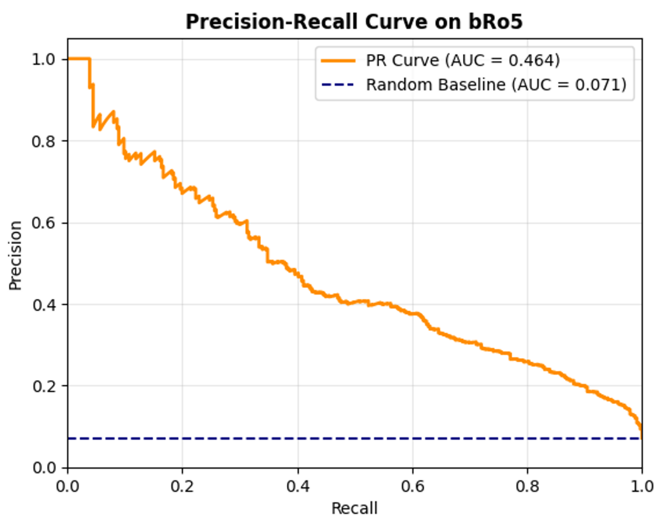


Fig. 5. Precision-Recall curve of the model trained on SangsterLogP.

6. However, since the overall predicted logP value is close to the Crippen method ground truth, it can be inferred that the model somehow simply flipped the contributions within the functional group without changing the values on other atoms, and thus the overall prediction.

In molecule 6, in the carbon with 3 bonded fluorine groups, the Crippen method heavily attributed a positive influence from each fluorine atom, while there is a slight negative influence on the carbon atom. On the other hand, LogPGNN attributed a slight positive influence on the fluorine atoms but a larger positive influence on the carbon atom. Thus, in these cases and some other cases not presented here, the model learned an alternate rule that is still consistent with the rules by the Crippen method, as seen by the very high R^2 shown in Figure 3.

2) SangsterLogP: Figure 7 shows the feature attribution (Crippen, integrated gradients, and SHAP) and edge attention weights of the LogPGNN model trained on the SangsterLogP dataset for various molecules.

For a sanity check, we see the feature attributions for simple organic compounds: benzene and phenol. For the first molecule, benzene in Figure 7, we see that all three attributions give a roughly equal and positive logP contribution to each carbon atom. In fact, the LogPGNN model predicted a logP very close to that of the Crippen method, both of which are a bit far from the actual logP of benzene. This means that the model was able to capture some of the Crippen rules from the SangsterLogP dataset, but due to the simplicity of the molecule, it defaulted to only the basic rules. The weight attention also gives equal importance to all bonds.

For the second molecule, phenol, all atom attribution methods captured the significant negative logP influence on the hydroxyl group. However, the Crippen method attributions differed from the LogPGNN attribution in how it distributed the positive influence of the logP throughout its six carbons. The Crippen method gives the most importance to the carbon where the hydroxyl group is attached (the ipso-carbon), whereas both model attribution gives the carbons opposite to the hydroxyl group (para-carbon and meta-carbons) more importance instead. It can be argued that the ipso-carbon loses its lipophilicity due to the electronegativity of the bonded oxygen, while the para-carbon and meta-carbons, due to their distance, retain most of their lipophilicity as it was still in benzene. The edge attentions show that the bond between the ipso-carbon and its attached hydroxyl group is the most crucial “information highway” to pass the information.

A random test set molecule shows an example where LogPGNN performed significantly better than the static Crippen method, which is way off. Although we have this logP discrepancy, the sign of the influence between Crippen and the model is similar, including the flipping between the

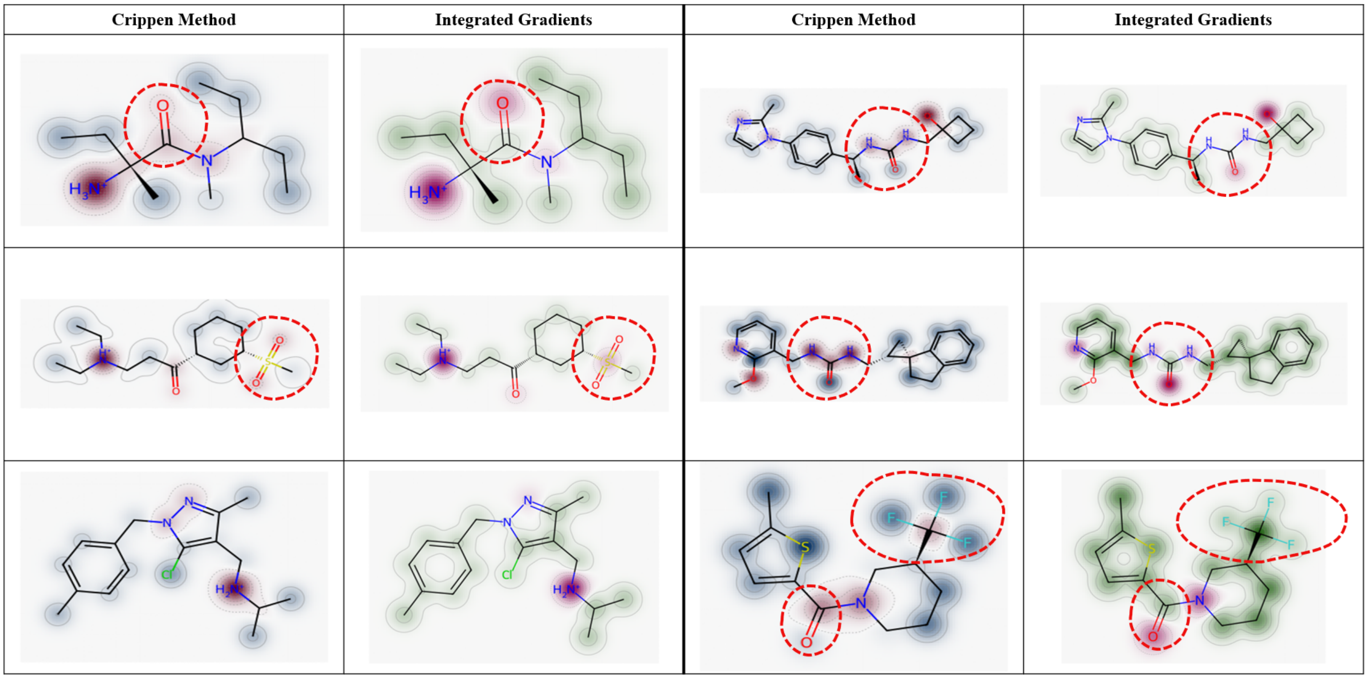


Fig. 6. Explainability of logP for model trained on ZINC 250k for random test molecules. Red circles indicate differences between attribution methods.

oxygen and carbon of the carbonyl group (C=O). Thus, we can attribute the actual magnitude of the influence as causing the difference.

In relation to the previous observation, also note that upon adding the atom attributions of both integrated gradients and SHAP, they unfortunately do not add exactly to the final logP, which violates the axiom of completeness for integrated gradients and local accuracy for SHAP. For the integrated gradients, this may be due to the imperfect "baseline" atom being used, which did not correspond to a predicted logP of 0 as needed for the effectiveness of integrated gradients [7]. Another reason is that the number of steps in the integrated gradients to approximate the fundamental integral used in the method might not be enough, although the delta of the numerical integrator is checked so that it does not exceed a threshold. For the SHAP attribution, note that a sampling version was used, which only uses a subset of all combinations and thus does not guarantee local accuracy [28].

Related to the sampling version of SHAP, do note that the attribution diagram generated is technically not deterministic since there was no set random seed used, but inspecting visually does not really show any noticeable difference.

Another test set molecule was examined, but in this case, both the Crippen and the model are way off, but at opposite extremes—Crippen predicted a higher logP while the model picked a lower logP. The situation with the trifluoromethyl group (-CF_3) that was noted earlier repeats in this case, which means that this alternative rule may be more consistent with actual chemistry. Also, another major difference between the Crippen method and both model attributes is in the oxygen in the ether groups (-O-). The Crippen method gives it a negative influence, whereas LogPGNN gives it a positive influence. On the other hand, for the oxygen on the furan ring (five-membered ring), the opposite happens, where Crippen gives it positive, while LogPGNN gives it negative. Overall, this resulted in the Crippen overestimating logP while the model underestimates.

For the last example, a molecule was chosen from the SAMPL6 challenge that has participants compete in having a low error prediction of the logP of 11 chosen molecules using various computational methods, including machine learning. Specifically, the molecule tagged as SM12 was chosen since it has a counterion in its SMILES string, which the challenge facilitators said should not change the logP of the molecule. Without including the counterion, the resulting logP model prediction is only slightly worse than that of Crippen. Looking at the bond attentions, it showed that the linking nitrogen between the two rings has a close attention weight, meaning that this nitrogen atom was a crucial information highway in passing the information between the two rings in the final logP prediction.

LogPGNN actually works with atoms with a counterion, although that counterion would be a node that does not have an edge connection to any nodes on the main molecular graph. Including the chlorine counterion in the molecular model makes all the attribution methods have a worse prediction. Especially for the LogPGNN model, it is seen that the model focused instead on the counterion while making the influence of the main molecule less significant. This means that in using this LogPGNN model, all counterions should be removed, especially if, chemically, it should not change the logP of the main molecule.

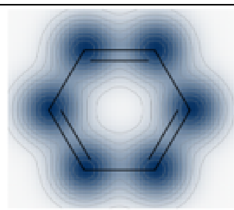
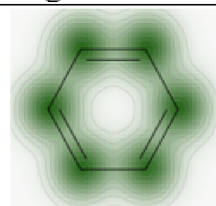
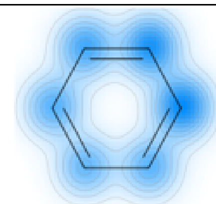
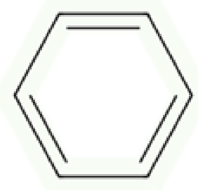
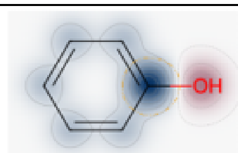
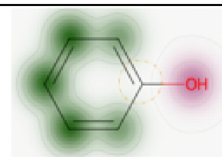
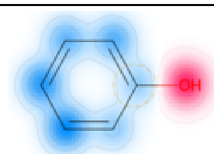
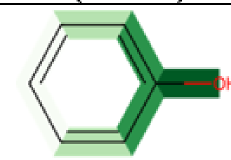
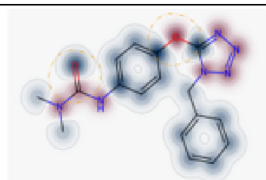
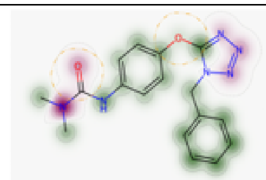
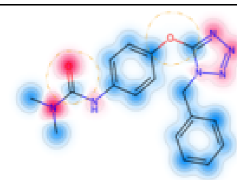
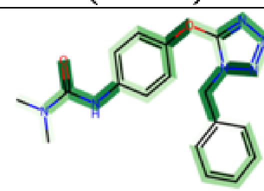
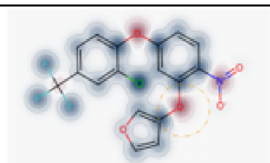
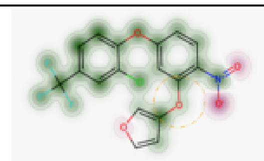
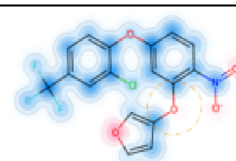
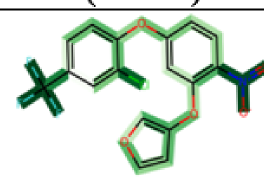
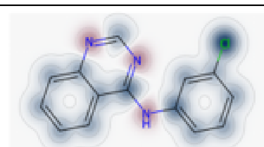
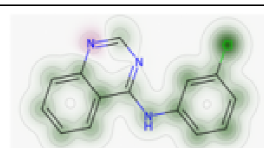
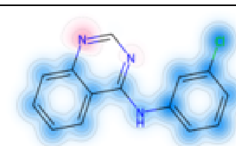
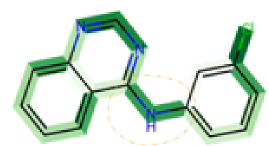
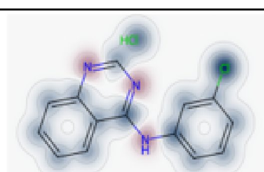
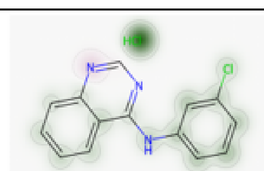
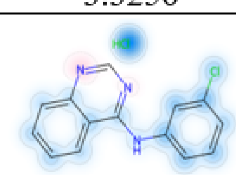
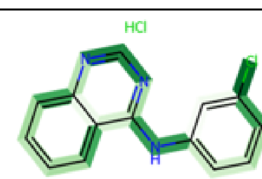
Name	Crippen method	Integrated gradients	SHAP	Attention
(1) Benzene	 1.6866	 1.6822	 1.6822	 **(2.1300)**
(2) Phenol	 1.3922	 1.2692	 1.2692	 **(1.4800)**
(3) Random Test Set Molecule #1	 2.6072	 1.7362	 1.7362	 **(1.7000)**
(4) Random Test Set Molecule #2	 6.4446	 3.6032	 3.6032	 **(4.3300)**
(5) SAMPL6 SM12	 4.0268	 3.3256	 3.3256	 **(3.8300)**
(6) SAMPL6 SM12 w/ Counterion	 4.4486	 2.6182	 2.6182	 **(3.8300)**

Fig. 7. Explainability of logP for model trained on SangsterLogP for chosen molecules along with the predicted logP values and actual logP value (bolded in the attention column).

Although the LogPGNN model did not have very high predictive metrics, the explainability demonstrated makes up for it, having an actual basis that researchers can see on where the model did right, and especially what it might have attributed wrong, and thus the model can be refined further based on this information.

V. Conclusion and Recommendations

A. Conclusion

In this study, the PRISM framework was successfully implemented for molecular property regression with corresponding explainability. Specifically, the use of the logP molecular property for property regression was successfully implemented using two datasets containing computational and especially experimental logP data. This model, which utilizes GATv2, is then called the LogPGNN model. This model, trained on the computational ZINC 250k dataset, showed very high regression and classification metrics (the latter with respect to passing Lipinski's Ro5 criteria of logP ¡ 5), whereas trained on the experimental SangsterLogP dataset with scaffold splitting, it showed mediocre metrics. The LogPGNN model also exhibited explainability via atom logP attribution through integrated gradients and SHAP, and also bond message passing importance through the attention mechanism. LogPGNN captured most of the rules stated by the static rule-based Crippen method and even more complex interactions due to its dynamic nature, leading to slightly better overall performance metrics.

Although the LogPGNN model trained on the experimental SangsterLogP dataset exhibited mediocre baseline metrics, its evaluation under a scaffold split demonstrates that the model retains a decent capacity for real-world virtual screening generalization. Furthermore, the integration of explainability methods provides a diagnostic mechanism to iteratively refine the architecture, as it allows researchers to pinpoint exactly where the model miscalculates individual atomic contributions. Overall, the PRISM framework proved successful for this specific logP prediction task, suggesting that this methodology can be effectively extended to other molecular property prediction pipelines crucial for computer-aided drug design.

B. Recommendations

As the model relied on DL models and especially involved the attention mechanism, one way to improve performance is to include more data points. Thus, hopefully, there would be more developments in creating ever larger experimental logP datasets. Also, other ANN models or even other machine learning models can help increase the performance and maybe even combat the overfitting observed.

Another approach is that instead of using the raw values of the attributions, the rank in terms of absolute magnitude is used instead. This means that the focus instead is to find out which atoms or functional groups have the most influence on logP, and thus, researchers can focus on these if they want to modify the logP of a molecule.

A metric that was not measured in the study was the accuracy of the atom logP attributions. Note that the Crippen method cannot be used as the real-world ground truth due to its static rule-based nature, whereas the experimental logP only has the ground truth for the final logP but not for the contributions of each atom. Thus, a more in-depth theoretical analysis should be performed to check whether each attribution is consistent with current theories in chemistry.

A notable limitation of the current study is that the empirical accuracy of the atom-level logP attributions was not quantitatively evaluated. It is important to highlight that rule-based frameworks, such as the Crippen method, cannot serve as an absolute ground truth due to their static, deterministic nature. Conversely, experimental datasets provide ground-truth values exclusively for the macroscopic logP of the whole molecule, lacking granular resolution for individual atomic contributions. Consequently, a more rigorous theoretical analysis must be conducted to verify whether the model's attributions align consistently with established chemical principles, such as electronic inductive effects, localized hydrophobicity, and even chameleonicity.

The long-term goal of this PRISM framework is to apply this to multi-property prediction and attribution. Note that it is relatively trivial to extend the code to work with multiple regression with other properties, such as QED and SAS, and to also separate their attributions. However, the most challenging thing is to find an empirical dataset where each molecule has all these properties and is of a decent size.

The long-term objective of the PRISM framework is its extension to simultaneous multi-property prediction and joint attribution. While expanding the underlying architecture to accommodate multi-output regression for multiple molecular descriptors, such as QED and SAS, is relatively straightforward, the primary bottleneck resides in data availability. The critical challenge lies in sourcing a curated empirical dataset of sufficient scale wherein each molecule is comprehensively annotated with all target properties, highlighting a common constraint in multi-task molecular deep learning.

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AI Disclosure

During the preparation of this work, the authors utilized **Gemini** to assist with code implementation and conceptual clarification, and **Grammarly** to refine phrasing, syntax, and grammatical accuracy.

After using this tool/service, the author reviewed and edited the content as needed and take full responsibility for the content of the assignment.

References

[1] I. Doytchinova, "Drug Design—Past, Present, Future," *Molecules*, vol. 27, no. 5, p. 1496, Feb. 2022, ISSN: 1420-3049. DOI: 10.3390/molecules27051496. [Online]. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC8911833/.

[2] M. Roney and M. F. F. Mohd Aluwi, "The importance of in-silico studies in drug discovery," *Intelligent Pharmacy*, vol. 2, no. 4, pp. 578–579, Aug. 2024, ISSN: 2949-866X. DOI: 10.1016/j.ipha.2024.01.010. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2949866X24000200.

[3] S. Duvaud, C. Gabella, F. Lisacek, H. Stockinger, V. Ioannidis, and C. Durinx, "Expasy, the Swiss Bioinformatics Resource Portal, as designed by its users," *Nucleic Acids Research*, vol. 49, no. W1, W216–W227, Jul. 2021, ISSN: 0305-1048. DOI: 10.1093/nar/gkab225. [Online]. Available: https://doi.org/10.1093/nar/gkab225.

[4] J. Jumper et al., "Highly accurate protein structure prediction with AlphaFold," en, *Nature*, vol. 596, no. 7873, pp. 583–589, Aug. 2021, ISSN: 1476-4687. DOI: 10.1038/s41586-021-03819-2. [Online]. Available: https://www.nature.com/articles/s41586-021-03819-2.

[5] Y. A. Qadri, S. Shaikh, K. Ahmad, I. Choi, S. W. Kim, and A. V. Vasilakos, "Explainable Artificial Intelligence: A Perspective on Drug Discovery," *Pharmaceutics*, vol. 17, no. 9, p. 1119, Aug. 2025, ISSN: 1999-4923. DOI: 10.3390/pharmaceutics17091119. [Online]. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC12472777/.

[6] S. Lundberg and S.-I. Lee, *A Unified Approach to Interpreting Model Predictions*, arXiv:1705.07874 [cs.AI], Nov. 2017. DOI: 10.48550/arXiv.1705.07874. [Online]. Available: http://arxiv.org/abs/1705.07874.

[7] M. Sundararajan, A. Taly, and Q. Yan, *Axiomatic Attribution for Deep Networks*, arXiv:1703.01365 [cs.LG], Jun. 2017. DOI: 10.48550/arXiv.1703.01365. [Online]. Available: http://arxiv.org/abs/1703.01365.

[8] Open Resources for Nursing (Open RN), "Chapter 1 Pharmacokinetics & Pharmacodynamics," en, in *Nursing Pharmacology [Internet]. 2nd edition*, K. Ernstmeyer and E. Christman, Eds., Chippewa Valley Technical College, 2023. [Online]. Available: https://www.ncbi.nlm.nih.gov/books/NBK595006/.

[9] S. Bhal, "LogP—Making Sense of the Value," en, [Online]. Available: https://www.acdlabs.com/wp-content/uploads/download/app/physchem/making_sense.pdf.

[10] J. Noble, *What is a Graph Neural Network — IBM*, en, Feb. 2025. [Online]. Available: https://www.ibm.com/think/topics/graph-neural-network.

[11] A. Vaswani et al., *Attention Is All You Need*, arXiv:1706.03762 [cs], Aug. 2023. DOI: 10.48550/arXiv.1706.03762. [Online]. Available: http://arxiv.org/abs/1706.03762.

[12] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio, *Graph Attention Networks*, arXiv:1710.10903 [stat.ML], Feb. 2018. DOI: 10.48550/arXiv.1710.10903. [Online]. Available: http://arxiv.org/abs/1710.10903.

[13] S. Brody, U. Alon, and E. Yahav, *How Attentive are Graph Attention Networks?* arXiv:2105.14491 [cs.LG], Jan. 2022. DOI: 10.48550/arXiv.2105.14491. [Online]. Available: http://arxiv.org/abs/2105.14491.

[14] L. Z. Benet, C. M. Hosey, O. Ursu, and T. I. Oprea, "BDDCS, the Rule of 5 and drugability," eng, *Advanced Drug Delivery Reviews*, vol. 101, pp. 89–98, Jun. 2016, ISSN: 1872-8294. DOI: 10.1016/j.addr.2016.05.007.

[15] J. J. Irwin, T. Sterling, M. M. Mysinger, E. S. Bolstad, and R. G. Coleman, "ZINC: A Free Tool to Discover Chemistry for Biology," *Journal of Chemical Information and Modeling*, vol. 52, no. 7, pp. 1757–1768, Jul. 2012, ISSN: 1549-9596. DOI: 10.1021/ci3001277. [Online]. Available: https://doi.org/10.1021/ci3001277.

[16] B. I. Tingle et al., "ZINC-22-A Free Multi-Billion-Scale Database of Tangible Compounds for Ligand Discovery," *Journal of Chemical Information and Modeling*, vol. 63, no. 4, pp. 1166–1176, Feb. 2023, ISSN: 1549-9596. DOI: 10.1021/acs.jcim.2c01253. [Online]. Available: https://doi.org/10.1021/acs.jcim.2c01253.

[17] M. Krenn, F. Häse, A. Nigam, P. Friederich, and A. Aspuru-Guzik, "Self-Referencing Embedded Strings (SELFIES): A 100% robust molecular string representation," *Machine Learning: Science and Technology*, vol. 1, no. 4, p. 045 024, Dec. 2020, arXiv:1905.13741 [cs], ISSN: 2632-2153. DOI: 10.1088/2632-2153/aba947. [Online]. Available: http://arxiv.org/abs/1905.13741.

[18] S. A. Wildman and G. M. Crippen, "Prediction of Physicochemical Parameters by Atomic Contributions," *Journal of Chemical Information and Computer Sciences*, vol. 39, no. 5, pp. 868–873, Sep. 1999, ISSN: 0095-2338. DOI: 10.1021/ci990307l. [Online]. Available: https://doi.org/10.1021/ci990307l.

[19] T. Cirino, G. Caron, G. Ermondi, L. Charochkina, and I. V. Tetko, "SangsterLogP - the largest publicly available dataset of logP values," en, *Scientific Data*, May 2026, ISSN: 2052-4463. DOI: 10.1038/s41597-026-07357-2. [Online]. Available: https://www.nature.com/articles/s41597-026-07357-2.

[20] Q. Guo, S. Hernandez-Hernandez, and P. J. Ballester, *Scaffold Splits Overestimate Virtual Screening Performance*, arXiv:2406.00873 [q-bio.QM], Jun. 2024. DOI: 10.48550/arXiv.2406.00873. [Online]. Available: http://arxiv.org/abs/2406.00873.

[21] *Scaffold_train_valid_test_split — scikit-fingerprints documentation*. [Online]. Available: https://scikit-fingerprints.readthedocs.io/latest/modules/generated/skfp.model_selection.scaffold_train_valid_test_split.html.

[22] *Rdkit: Open-source cheminformatics*. [Online]. Available: https://www.rdkit.org.

[23] M. Fey and J. E. Lenssen, *Fast Graph Representation Learning with PyTorch Geometric*, en, Mar. 2019. [Online]. Available: https://arxiv.org/abs/1903.02428v3.

[24] A. Paszke et al., *PyTorch: An Imperative Style, High-Performance Deep Learning Library*, en, Dec. 2019. [Online]. Available: https://arxiv.org/abs/1912.01703v1.

[25] T. K. Rusch, M. M. Bronstein, and S. Mishra, "A Survey on Oversmoothing in Graph Neural Networks," 2023, Version Number: 1. DOI: 10.48550/ARXIV.2303.10993. [Online]. Available: https://arxiv.org/abs/2303.10993.

[26] I. Loshchilov and F. Hutter, "Decoupled Weight Decay Regularization," en, Sep. 2018. [Online]. Available: https://openreview.net/forum?id=Bkg6RiCqY7.

[27] T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama, *Optuna: A Next-generation Hyperparameter Optimization Framework*, en, Jul. 2019. [Online]. Available: https://arxiv.org/abs/1907.10902v1.

[28] N. Kokhlikyan et al., *Captum: A unified and generic model interpretability library for PyTorch*, en, Sep. 2020. [Online]. Available: https://arxiv.org/abs/2009.07896v1.

[29] G. M. Sullivan and R. Feinn, "Using Effect Size—or Why the P Value Is Not Enough," *Journal of Graduate Medical Education*, vol. 4, no. 3, pp. 279–282, Sep. 2012, ISSN: 1949-8349. DOI: 10.4300/JGME-D-12-00156.1. [Online]. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC3444174/.

[30] M. Egbert, A. Whitty, G. M. Keserű, and S. Vajda, "Why some targets benefit from beyond rule of five drugs," *Journal of medicinal chemistry*, vol. 62, no. 22, pp. 10 005–10 025, Nov. 2019, ISSN: 0022-2623. DOI: 10.1021/acs.jmedchem.8b01732. [Online]. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC7102492/.

[31] T. Tashima, "Emerging cellular uptake mechanisms of bRo5 PROTACs," *RSC Medicinal Chemistry*, Apr. 2026. DOI: 10.1039/D6MD00243A.

[32] R. Cao et al., "Enhanced Bioactivity of Natural Products by Halogenation: A Database Survey and Quantum Chemistry Calculation Study," *Journal of Medicinal Chemistry*, vol. 68, no. 10, pp. 10 486–10 496, May 2025, ISSN: 0022-2623. DOI: 10.1021/acs.jmedchem.5c00944. [Online]. Available: https://doi.org/10.1021/acs.jmedchem.5c00944.